

CIS 25 Midterm

Ticket System for Concert

Explanation

- This is a ticket management system for a concert
- You can only manage one concert at a time
- When starting out, you can either create a new concert or load one from a file
- Each concert has a name, standard ticket price, VIP ticket price, number of available tickets, total revenue, and ticket list
- As you sell tickets, they are saved in the program
 - With these you can see your guests
 - How many tickets you've sold
 - Search for a guest based off ticket id
 - They add to total revenue
- When you close the program it loads back to the file
 - The file is fully loaded from memory every time the program opens

Concepts

1. Data types
 - a. This program uses int, double, string, iostream, vector, and fstream datatypes
2. File types and I/O
 - a. This program loads data from a text file on opening
 - b. Then saves data back to the text file when closing
3. Pointers
 - a. The main object ConcertMgr has a vector of Ticket pointers
 - b. In main, ConcertMgr is created in heap memory and referenced by a pointer because I needed to construct it within an if statement
4. Arrays
 - a. The main object ConcertMgr has a vector of Ticket pointers
5. Binary Search
 - a. Users can search for guests by ticket ID
 - b. ConcertMgr does a binary search on the vector of tickets
6. Strings
 - a. Strings are used to save guest name in tickets, and concert name in ConcertMgr, among other places
7. Class interaction
 - a. ConcertMgr contains a vector of Ticket pointers
 - b. Ticket is a pure virtual class has subclasses of StandardTicket and VipTicket

Challenges

1. The biggest challenge I faced was time. I had a bit of trouble estimating how long each piece would take
2. Moreover, I struggled with classes that have a many to many relationship. At first I was going to have events and guests each as their own class. Then they would be bound together by tickets. However, I couldn't figure out where to start with this design so I simplified it to being a ticket manager for one concert

Program Execution

1. Welcome screen

```
C:\Users\abarr\Documents\cs X + v
Welcome to Concert Manager!
Here you can create or load a concert and manage the tickets for it
When numbered prompts pop up, please enter the number of the option you would like to select
Would you like to:
    1. Load current concert
    2. Create a new concert
    3. Exit program
```

2. If you load a concert that doesn't exist, you get error handling

```
Selection: 1
There is no current concert, please select again

Would you like to:
    1. Load current concert
    2. Create a new concert
    3. Exit program

Selection:
```

3. Start out by creating a new project

```
Would you like to:
    1. Load current concert
    2. Create a new concert
    3. Exit program

Selection: 2
Please enter concert details
Name: Music Festival
Standard ticket price: 10.00
VIP ticket price: 20.00
Total available tickets: 100
```

4. This is the homepage of concert manager which lets you choose from the following

```
Homepage
Would you like to:
    1. Sell tickets
    2. Search for guest
    3. Print guest list
    4. See concert details
    5. See box office details
    6. Exit program

Selection:
```

5. Lets start by selling a ticket

```
Selection: 1
Please enter ticket details
Guest Name: Jerry Seinfeld
Ticket Type:
    1. Standard
    2. VIP

Selection: 1
Quantity: 5

Ticket sold!
```

6. At this point I've added George and Elaine, now let's search by id. It will also say that the guest wasn't found if they are not on the list

```
Please enter the ticket ID of the guest you would like to search for
Ticket ID: 3

Elaine Benes, 6 Standard Tickets
```

7. Let's now close the program

```
Selection: 6

Thank you for using Concert Manager, goodbye
```

8. I will now open my loaded concert and check the guest list

```
Guests List:
Jerry Seinfeld, Standard
George Costanza, Standard
Elaine Benes, Standard
```

9. We can view the concert details we entered

```
Tonight's Event: Music Festival  
Standard Tickets: $10.00  
VIP Tickets: $20.00
```

10. We can see our box office

```
Revenue: $30.00  
Tickets Sold: 14  
Remaining Tickets: 36
```

11. Also, if we enter something incorrectly, there is an error check

```
Selection: 7  
  
Please choose a valid option
```